PROBLEM 28

Spherical Pendulum with a Vertically Oscillating Suspension Point

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Translating Frame + Inertia Force

A point particle P of mass m is attached with a massless cord of length L to the point A, as shown in the figure. The point A is oscillating vertically with respect to the fixed reference frame $Oxyz$ according to the law $z = a\sin\omega t$, where a and ω are known constants. The position of the particle P with respect to the moving reference frame $Ax'y'z'$ is completely specified by the angles φ and θ shown in the figure.

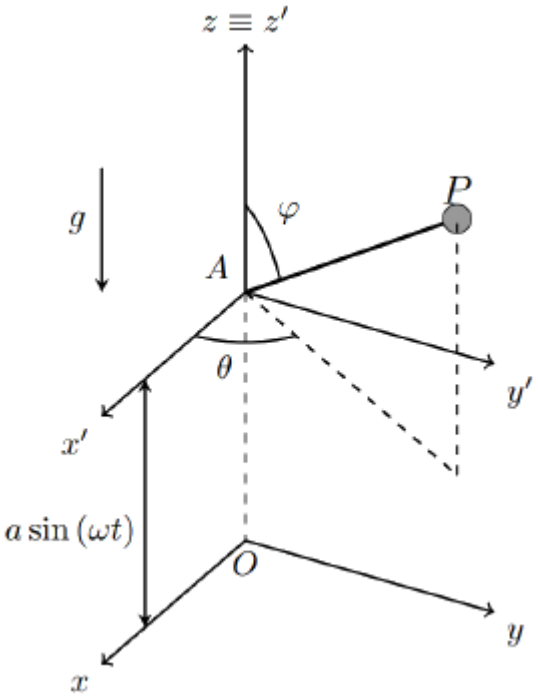


Figure: Point A oscillating vertically as a sin(ωt) above O; translating axes Ax'y'z'; particle P on a cord from A, located by the angle φ from Az' and the azimuth θ.

At $t = 0$ the relative position and relative velocity of the particle are defined by the values

$$\theta_0 = 0, \qquad \varphi_0 = 30^\circ, \qquad \dot{\theta}_0 = \sqrt{\frac{2g}{\sqrt{3}L}}, \qquad \dot{\varphi}_0 = 0,$$

where g is the gravitational acceleration. We define the following unit vectors that define a spherical coordinate system

$$\mathbf{e}_r = \sin\varphi\cos\theta\,\mathbf{i} + \sin\varphi\sin\theta\,\mathbf{j} + \cos\varphi\,\mathbf{k}, \qquad \mathbf{e}_\varphi = \cos\varphi\cos\theta\,\mathbf{i} + \cos\varphi\sin\theta\,\mathbf{j} - \sin\varphi\,\mathbf{k}, \qquad \mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}.$$

The partial derivatives of these vectors with respect to φ and θ are

$$\frac{\partial \mathbf{e}_r}{\partial \varphi} = \mathbf{e}_\varphi, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \varphi} = -\mathbf{e}_r, \quad \frac{\partial \mathbf{e}_\theta}{\partial \varphi} = \mathbf{0}, \quad \frac{\partial \mathbf{e}_r}{\partial \theta} = \sin\varphi\,\mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \theta} = \cos\varphi\,\mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\theta}{\partial \theta} = -\sin\varphi\,\mathbf{e}_r - \cos\varphi\,\mathbf{e}_\varphi.$$

(a) Show that the relative acceleration of P with respect to $Ax'y'z'$ can be expressed as $\mathbf{a}'_P = -L\left(\dot{\varphi}^2 + \dot{\theta}^2\sin^2\varphi\right)\mathbf{e}_r + L\left(\ddot{\varphi} - \dot{\theta}^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + L\left(\ddot{\theta}\sin\varphi + 2\dot{\theta}\dot{\varphi}\cos\varphi\right)\mathbf{e}_\theta$.

(b) Write down a system of three ordinary differential equations that once solved would yield the angles θ and φ, and the tension on the cord as a function of time (you do not need to solve it).

(c) At $t = 0$, determine the relative acceleration of P with respect to $Ax'y'z'$.

(d) At $t = 0$, determine the value of the tension on the cord.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a particle P (mass m) hangs from a massless cord of length L fixed at A; A moves on the vertical axis with $z_A = a\sin\omega t$. In the frame $Ax'y'z'$ the particle is located by the spherical angles φ (from the upward axis Az' to AP, figure) and θ (azimuth from Ax').
- **Hypotheses:** $Oxyz$ is inertial with Oz upwards and gravity $-g\,\mathbf{k}$; the cord is massless and taut, so it acts on P with a tension $\mathbf{T} = -T\,\mathbf{e}_r, T \geq 0$, directed from P to A; no friction or drag. Here ω is the frequency of the oscillation of A, not an angular velocity of the frame.
- **Why a moving frame:** in $Ax'y'z'$ the cord has fixed length, so P moves on a sphere of radius L described by two angles. The frame does not rotate, so the only inertia force is the one due to the acceleration of its origin.
- **Degrees of freedom:** 3 coordinates minus the constraint $|\mathbf{AP}| = L$: CDOF $= 2$, generalized coordinates (φ, θ). The motion of A is prescribed.
- **Initial conditions:** $\theta_0 = 0, \varphi_0 = 30^\circ = \pi/6, \dot\theta_0 = \sqrt{2g/(\sqrt{3}L)}, \dot\varphi_0 = 0$ (relative values).

Symbol	Meaning	SI unit
m	mass of P	kg
L	cord length	m
a	amplitude of the motion of A	m
ω	frequency of the motion of A	rad/s
g	gravitational acceleration	m/s^2
φ, θ	polar and azimuthal angles of AP in $Ax'y'z'$	rad
T	cord tension	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = Oxyz = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Moving frame:** $S_1 = Ax'y'z' = \{A; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$: its axes stay parallel to those of $Oxyz$ (figure), so the frame only translates and $[{}_0R_1] = I$ ($\det I = 1$, $II^T = I$). In the notation of the theory, $\mathbf{a}'_P \equiv \mathbf{a}_1^P$.
- **Motion of S_1: $\boldsymbol{\omega}_{10} = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$** and $\mathbf{r}_0^A = a \sin \omega t \, \mathbf{k}$. By the chain rule, $\frac{d}{dt} \sin \omega t = \omega \cos \omega t$ and $\frac{d}{dt} \cos \omega t = -\omega \sin \omega t$:

$$\mathbf{v}_0^A = a\omega \cos \omega t \, \mathbf{k}, \qquad \mathbf{a}_0^A = -a\omega^2 \sin \omega t \, \mathbf{k}$$

- **Spherical basis $\{\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{e}_\theta\}$** of the statement. The matrix with these columns (components in $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$)

$$[{}_1R_s] = \begin{pmatrix} \sin\varphi\cos\theta & \cos\varphi\cos\theta & -\sin\theta \\ \sin\varphi\sin\theta & \cos\varphi\sin\theta & \cos\theta \\ \cos\varphi & -\sin\varphi & 0 \end{pmatrix}$$

has orthonormal columns ($|\mathbf{e}_r|^2 = \sin^2\varphi(\cos^2\theta + \sin^2\theta) + \cos^2\varphi = 1$, $\mathbf{e}_r \cdot \mathbf{e}_\varphi = \sin\varphi\cos\varphi - \cos\varphi\sin\varphi = 0$, $\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$, $\mathbf{e}_\varphi \cdot \mathbf{e}_\theta = 0$), so $[{}_1R_s]^T[{}_1R_s] = I$; and $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{e}_\theta$, so $\det[{}_1R_s] = 1$ (right-handed). The last row gives the vertical: $\mathbf{k} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$.

- **Relative position:** $\mathbf{r}_1^P = L\,\mathbf{e}_r$. Since $\varphi < \pi/2$ means $\mathbf{e}_r \cdot \mathbf{k} > 0$, at $t = 0$ ($\varphi_0 = 30^\circ$) the particle is **above** A, as drawn in the figure.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative acceleration

Pedagogical Justification: $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are constant in S_1; the spherical vectors depend on time only through $\varphi(t)$ and $\theta(t)$, so their derivative in S_1 follows from the multivariable chain rule $\dfrac{d\mathbf{e}}{dt} = \dfrac{\partial \mathbf{e}}{\partial \varphi}\dot{\varphi} + \dfrac{\partial \mathbf{e}}{\partial \theta}\dot{\theta}$ with the partial derivatives of the statement.

Time derivatives of the basis (in S_1):

$$\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r + \dot\theta\cos\varphi\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\dot\theta\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi)$$

Relative velocity (L constant):

$$\mathbf{v}_1^P = L\,\dot{\mathbf{e}}_r = L\dot\varphi\,\mathbf{e}_\varphi + L\dot\theta\sin\varphi\,\mathbf{e}_\theta$$

Relative acceleration. Product rule on each term: $\frac{d}{dt}\left(L\dot\varphi\,\mathbf{e}_\varphi\right) = L\ddot\varphi\,\mathbf{e}_\varphi + L\dot\varphi\,\dot{\mathbf{e}}_\varphi$ and $\frac{d}{dt}\left(L\dot\theta\sin\varphi\,\mathbf{e}_\theta\right) = L\ddot\theta\sin\varphi\,\mathbf{e}_\theta + L\dot\theta\cos\varphi\,\dot\varphi\,\mathbf{e}_\theta + L\dot\theta\sin\varphi\,\dot{\mathbf{e}}_\theta$ (chain rule $\frac{d}{dt}\sin\varphi = \cos\varphi\,\dot\varphi$). Substituting the basis derivatives:

$$\mathbf{a}_1^P = L\ddot\varphi\,\mathbf{e}_\varphi + L\dot\varphi\left(-\dot\varphi\,\mathbf{e}_r + \dot\theta\cos\varphi\,\mathbf{e}_\theta\right) + L\ddot\theta\sin\varphi\,\mathbf{e}_\theta + L\dot\theta\dot\varphi\cos\varphi\,\mathbf{e}_\theta - L\dot\theta^2\sin\varphi\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi)$$

Collecting along $\mathbf{e}_r$ ($-L\dot\varphi^2 - L\dot\theta^2\sin^2\varphi$), $\mathbf{e}_\varphi$ ($L\ddot\varphi - L\dot\theta^2\sin\varphi\cos\varphi$) and $\mathbf{e}_\theta$ ($L\dot\varphi\dot\theta\cos\varphi + L\ddot\theta\sin\varphi + L\dot\theta\dot\varphi\cos\varphi$):

$$\mathbf{a}'_P \equiv \mathbf{a}_1^P = -L\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right)\mathbf{e}_r + L\left(\ddot\varphi - \dot\theta^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + L\left(\ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi\right)\mathbf{e}_\theta$$

which is the expression of the statement.

(b) Equations of motion and tension

> **Pedagogical Justification:** in the translating frame, Notes Eq. 4.63 with $\boldsymbol{\omega}_{10} = \boldsymbol{\alpha}_{10} = \mathbf{0}$ reduces to $m\mathbf{a}_1^P = \mathbf{F} - m\mathbf{a}_0^A$ (equivalently Notes Eq. 3.29, $\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P$, inserted in $m\mathbf{a}_0^P = \mathbf{F}$). Projecting on the spherical basis separates the tension (only along $\mathbf{e}_r$) from the two equations of motion.

Forces in S_1. Tension $-T\,\mathbf{e}_r$; weight $-mg\,\mathbf{k}$; inertia force of the origin $\mathbf{F}_A = -m\mathbf{a}_0^A = ma\omega^2\sin\omega t\,\mathbf{k}$. Weight and $\mathbf{F}_A$ combine into an effective, time-dependent gravity:

$$-mg\,\mathbf{k} + ma\omega^2 \sin\omega t\,\mathbf{k} = -m\,g_e(t)\,\mathbf{k}, \qquad g_e(t) = g - a\omega^2 \sin\omega t$$

With $\mathbf{k} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$, this force is $-mg_e\cos\varphi\,\mathbf{e}_r + mg_e\sin\varphi\,\mathbf{e}_\varphi$.

Three components of $m\mathbf{a}_1^P = -T\,\mathbf{e}_r - mg_e\,\mathbf{k}$:

$$\mathbf{e}_r: \quad -mL\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right) = -T - mg_e(t)\cos\varphi$$

$$\mathbf{e}_\varphi: \quad mL\left(\ddot\varphi - \dot\theta^2\sin\varphi\cos\varphi\right) = mg_e(t)\sin\varphi$$

$$\mathbf{e}_\theta: \quad mL\left(\ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi\right) = 0$$

The requested system (solved for the unknowns):

$$\begin{cases} \ddot\varphi = \dot\theta^2\sin\varphi\cos\varphi + \dfrac{g - a\omega^2\sin\omega t}{L}\sin\varphi \\ \ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi = 0 \\ T = mL\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right) - m\left(g - a\omega^2\sin\omega t\right)\cos\varphi \end{cases}$$

The first two equations, with the initial conditions, give $\varphi(t)$ and $\theta(t)$; the third gives $T(t)$. Multiplying the second by $\sin\varphi$, product and chain rules show it is $\frac{d}{dt}\left(\dot\theta\sin^2\varphi\right) = \ddot\theta\sin^2\varphi + 2\dot\theta\sin\varphi\cos\varphi\,\dot\varphi = 0$: the vertical component of the angular momentum about A, $mL^2\dot\theta\sin^2\varphi$, is conserved, because tension, weight and $\mathbf{F}_A$ have no moment about the vertical axis through A (Topic 4).

(c) Relative acceleration at $t = 0$

Data: $\sin\omega \cdot 0 = 0$, so $g_e(0) = g$; $\sin\varphi_0 = \frac{1}{2}$, $\cos\varphi_0 = \frac{\sqrt{3}}{2}$, $\dot\theta_0^2 = \dfrac{2g}{\sqrt{3}L}$, $\dot\varphi_0 = 0$.

Angular accelerations. From the first equation of (b):

$$\ddot\varphi_0 = \frac{2g}{\sqrt{3}L}\cdot\frac{1}{2}\cdot\frac{\sqrt{3}}{2} + \frac{g}{L}\cdot\frac{1}{2} = \frac{g}{2L} + \frac{g}{2L} = \frac{g}{L}$$

From the second, $\ddot\theta_0 \sin\varphi_0 = -2\dot\theta_0 \cdot 0 \cdot \cos\varphi_0 = 0$, so $\ddot\theta_0 = 0$.

Components of $\mathbf{a}_1^P(0)$. $\dot\theta_0^2 \sin^2\varphi_0 = \dfrac{2g}{\sqrt{3}L}\cdot\dfrac{1}{4} = \dfrac{g}{2\sqrt{3}L}$ and $\dot\theta_0^2 \sin\varphi_0\cos\varphi_0 = \dfrac{2g}{\sqrt{3}L}\cdot\dfrac{\sqrt{3}}{4} = \dfrac{g}{2L}$:

$$\mathbf{a}_1^P(0) = -\frac{g}{2\sqrt{3}}\,\mathbf{e}_r + L\left(\frac{g}{L} - \frac{g}{2L}\right)\mathbf{e}_\varphi + 0\,\mathbf{e}_\theta = -\frac{g}{2\sqrt{3}}\,\mathbf{e}_r + \frac{g}{2}\,\mathbf{e}_\varphi$$

At $\theta = 0$, $\varphi = 30^\circ$: $\mathbf{e}_r = \frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{k}$ and $\mathbf{e}_\varphi = \frac{\sqrt{3}}{2}\mathbf{i} - \frac{1}{2}\mathbf{k}$, so the $\mathbf{i}$ component is $-\frac{g}{4\sqrt{3}} + \frac{\sqrt{3}g}{4} = \frac{(3-1)g}{4\sqrt{3}} = \frac{g}{2\sqrt{3}}$ and the $\mathbf{k}$ component is $-\frac{g}{4} - \frac{g}{4} = -\frac{g}{2}$:

$$\mathbf{a}_1^P(0) = \frac{g}{2\sqrt{3}}\,\mathbf{i} - \frac{g}{2}\,\mathbf{k}, \qquad \left|\mathbf{a}_1^P(0)\right| = g\sqrt{\frac{1}{12} + \frac{1}{4}} = \frac{g}{\sqrt{3}}$$

(d) Tension at $t = 0$

From the third equation of (b):

$$T(0) = mL\left(0 + \frac{g}{2\sqrt{3}L}\right) - mg\,\frac{\sqrt{3}}{2} = mg\left(\frac{1}{2\sqrt{3}} - \frac{3}{2\sqrt{3}}\right) = -\frac{mg}{\sqrt{3}}$$

Interpretation. The value is negative: at $t = 0$ the cord would have to push the particle outwards. A cord can only pull, so with the data as stated the cord is not taut at $t = 0$ and the taut-cord model of (a)–(c) does not apply. The actual motion starts as a free motion (the cord

goes slack), with $m\mathbf{a}_1^P = -mg_e\,\mathbf{k}$, i.e. $\mathbf{a}_1^P(0) = -g\,\mathbf{k}$ instead of the value of (c). The result $T(0) = -mg/\sqrt{3}$ would be meaningful as a compression if P were held by a massless rigid rod instead of a cord.

Data inconsistency in the statement (no official key)
With φ measured from the upward axis Az' (figure and $\mathbf{e}_r \cdot \mathbf{k} = \cos\varphi$), the given $\dot\theta_0$ cannot keep the cord taut: $T(0) = -mg/\sqrt{3} < 0$. The given value $\dot\theta_0^2 = 2g/(\sqrt{3}L) = g/(L\cos 30^\circ)$ is exactly the rate of a steady **conical pendulum hanging below** A at 30° from the downward vertical. Under that alternative reading (which contradicts the figure and the given $\mathbf{e}_r$, so it is only a remark), the same method gives $\ddot\varphi_0 = 0$, $\mathbf{a}_1^P(0) = -\dfrac{g}{\sqrt{3}}\,\mathbf{i}$ (purely centripetal, towards the axis) and $T(0) = mL\dot\theta_0^2\sin^2\varphi_0 + mg\cos\varphi_0 = \dfrac{mg}{2\sqrt{3}} + \dfrac{\sqrt{3}mg}{2} = \dfrac{2mg}{\sqrt{3}}$. The answers above follow the statement as written.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Effective gravity.** In the translating frame of A the oscillation of the support only modulates gravity, $g_e(t) = g - a\omega^2\sin\omega t$ (Slide 28: the force due to the acceleration of the origin is indistinguishable from a gravitational field). When $a\omega^2 > g$, g_e changes sign periodically and the upper position can be dynamically stabilised (inverted pendulum with a vibrating support; extension, not course material).
- **Limits.** $a \to 0$ or $\omega \to 0$: $g_e = g$ and the system is the classical spherical pendulum written with φ from the upward vertical (equilibrium $\varphi = 0$ unstable, $\varphi = \pi$ stable). $\dot\theta = 0$: the θ equation is satisfied identically and the motion is planar.
- **Signs at $t = 0$.** $\ddot\varphi_0 = g/L > 0$: the particle starts to fall away from the upward vertical, since both the weight and the centrifugal effect of $\dot\theta_0$ push φ to grow.
- **Dimensions:** $[a\omega^2] = \mathrm{m}\cdot\mathrm{s}^{-2} = [g]$; $[\dot\theta^2] = [g/L] = \mathrm{s}^{-2}$, so each term of the $\ddot\varphi$ equation is in s^{-2}; $[mL\dot\theta^2] = [mg] = \mathrm{N}$; $[L\dot\varphi^2] = [L\ddot\theta] = \mathrm{m}\cdot\mathrm{s}^{-2}$.

- **Symbolic check (sympy, `verify_P28.py`):** orthonormality and handedness of the spherical basis and the six partial derivatives of the statement; $\mathbf{v}_1^P$ and $\mathbf{a}_1^P$ by direct differentiation; $\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P$; the three equations of (b) solved from the vector equation; $\ddot{\varphi}_0 = g/L, \ddot{\theta}_0 = 0, \mathbf{a}_1^P(0), |\mathbf{a}_1^P(0)| = g/\sqrt{3}, T(0) = -mg/\sqrt{3}$; and the alternative reading ($\ddot{\varphi}_0 = 0, T(0) = 2mg/\sqrt{3}$): all verified.

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